MathForces

Student ID: _____ **Time: 120 minutes**

Student Name: _____ **Total Marks: 180**

Instructions: *Answer all questions in the space provided below each question. Write clearly.* ***Each question carries 10 marks****.* ***Marks for sub-parts, if any, are shown beside them****.*

Main Section (16 × 10)

Q1. Bo loves huge numbers, while Jack enjoys asking tricky mathematical questions. One day, Jack challenged Bo with an integer n and asked -How many trailing zeros are there in **n!** ?
Recall that, n! = 1 x 2 x 3 x ... x n.
Trailing zeros are the **zeros** that **appear at the end** of the decimal representation of a number. For better understanding, 2500 has two trailing zero and 25001 has no trailing zero.
Since Bo is busy, help him answer Jack's following questions.
(i). How many tailing zeros are there in 13! ? **[2]**
(ii). How many tailing zeros are there in 21! ? **[4]**
(iii). How many tailing zeros are there in 29! ? **[4]**

Answer:

Q2. A frog starts on **Stone 1** . In each jump, it may move **either 3 stones or 4 stones** forward. Answer the following questions: [Yes\No]

(i)Can the frog reach Stone 6? If yes, what is the minimum number of jumps required? **[3]**

(ii) Can the frog reach Stone 17? If yes, what is the minimum number of jumps required? **[7]**

Answer:

Q3. How many different shortest paths are there from point A(0,0) to point B(8,8) on a grid, if you can only move right or up?

Answer:

Q4. The annual Grand Festival is about to begin, and the organizers have three important tasks to complete: Decorating the venue (Task A), Preparing the food stalls (Task B), Setting up the stage (Task C).There are X different ways to complete Task A, Y different ways to complete Task B, and Z different ways to complete the task C.

A complete festival plan consists of choosing exactly one way for each of the three tasks. Determine how many distinct festival plans can be created.

(i).Where, X=2 , Y=8, Z=1 **[5]**

(ii).Where, X=2, Y=5 , Z=7 **[5]**

Answer:

Q5. The king is arranging a grand parade with soldiers numbered from 1 to N. He wants them tostand in a single row forming a permutation of the numbers 1 to N. A permutation is anarrangement of all numbers from 1 to N in which every number appears exactly once.The parade is balanced if the sum of the last half of the row is exactly half of the sum of all elements. Write a valid permutation [Multiple solution is possible]. If such an arrangement is impossible, write "NO".

i. where, N=16 **[4]**

ii. where, N=10 **[3]**

iii. where, N=12 **[3]**

Answer:

Q6. A school has 18 lockers, numbered from 1 to 18. Initially, all lockers are closed.

Then, 18 students walk past the lockers one after another.

. The 1st student toggles every locker (opens a closed locker and closes an open locker).

. The 2nd student toggles every 2nd locker (2, 4, 6, 8, ...).

. The 3rd student toggles every 3rd locker (3, 6, 9, 12, ...).

. This continues in the same way.

. Finally, the 18th student toggles only locker 18.

After all 18 students have finished, determine how many lockers remain open.

Answer:

Q7. Suppose you live in a magical world where magic is a part of everyday life. One day, after returning home from your daily chores, you discover that your house has been sealed by a magical lock. The lock consists of the numbers 1 through 7 arranged in a circle. A magical arrow initially points to 3. You also find an inscription on your door stating that 76 moves are required to break the lock.

At each move, the arrow rotates clockwise by a number of positions equal to the number it is currently pointing at.

For example, 3 -> 6 -> 5 -> 3 -> …

This process continues until all 76 moves have been made.

Your task is to determine which number the arrow will point to after the final move, allowing you to unlock your house.

Answer:

Q8. Yamal and Kylian are trapped in a mysterious world of numbers. To find their way back home, they must solve a mathematical puzzle. You are given a number N. Your task is to determine its largest prime factor. N= 98^150 + 98^151 + 98^152

Answer:

Q9. Suppose you are playing in the FIFA World Cup 2026. The match has gone all the way to a penalty shootout, and you step up to take the first kick for your team. Your first attempt carries a 25 percent chance of beating the goalkeeper. If you miss, the miss only sharpens your focus for the next kick, and your chance of scoring rises by 5 percentage points with every failed attempt.

Determine whether there exists an attempt number k such that, if you have missed all previous k−1 attempts, you will surely score (100%) on the k'th attempt. If such an attempt exists, output its number; otherwise, determine that it is impossible.

Answer:

Q10. For over a century, every royal decree issued by the Empire was assigned a unique page number, beginning with 1. The scribes carefully wrote every page number in order, one after another, using ink that could never be erased. When the final decree was completed, the Chief Scribe counted the total number of digits used to write all page numbers from 1 up to the final page. The count was exactly 48,889. The Emperor then engraved only the final page number onto a golden seal, believing that anyone intelligent enough to recover it from this single clue would be worthy of entering the royal archives. What number was engraved on the Emperor's golden seal?

Answer:

Q11. Mr. White and Mr. Pinkman own a famous ice factory known for the purity of its products. To produce one block of ICE, they use exactly X doses of chemicals, where X is the total number of doses in the ICE formula. Each dose is either Hydrogen (H) or Oxygen (O).Due to a special chemical rule, two Oxygen doses may never appear consecutively; that is, they cannot add two or more Oxygen doses in a row.

For example, HHH, HO, O, HOHHHO are valid ICE formulas. Whereas OHOOOH,

HOOHHO are not valid. when X = 2, there are 3 valid ICE formulas: HH, HO, and OH.

(i) If X = 4, how many different valid ICE formulas can be created? **[3]**

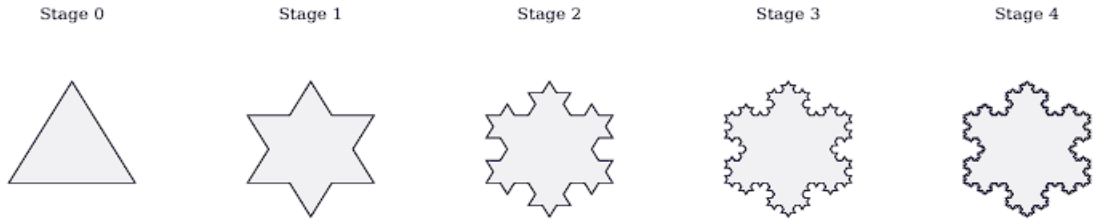
(ii If X = 13, how many different valid ICE formulas can be created? **[7]**

Answer:

Q12. There are 25 people standing in a circle. Starting from person 0, you count every 3rd person and eliminate them. The counting continues from the next person after the eliminated one, and you keep going around the circle until only 1 person remains. Who is the survivor?

Answer:

Q13.



A Koch snowflake begins as an equilateral triangle (Stage 0) with side length $l_0 = 18.0$ cm. At each stage every edge is divided into three equal parts and the middle part is replaced by two sides of a new outward-pointing equilateral triangle.

Formulate an expression for the total area A_n of the snowflake at any stage n, as a summation in terms of A_0 (the area of the initial triangle) and n. Hence evaluate the total area as $n \to \infty$.

Answer:

Q14. A positive integer is called a Shundor Number if it satisfies the following conditions:

1.The number is a palindrome.

2.Every proper divisor of the number that is greater than 1 is a prime palindrome greater than 11.

Recall that:

A palindrome is a number that reads the same forwards and backwards. For example, 121, 1331, and 101 are palindromes, while 123 is not.

A proper divisor of a positive integer is any positive divisor smaller than the number itself. For example, the proper divisors of 12 are 1, 2, 3 and 4.

A prime palindrome is a number that is both prime and a palindrome. For example, (101) is a prime palindrome.

For example, 10201 is a Shundor Number because its only proper divisor greater than 1 is 101, which is a prime palindrome greater than 11.

If N=12321, determine the smallest Shundor Number that is strictly greater than N.

Answer:

Q15.

Find the number of integer solutions $(x_1, x_2, \ldots, x_{2028})$ to the equation

$$1 + \sum_{i=1}^{2028} x_i = 0,$$

subject to the condition that $x_i \geq -1014$ for all integers i such that $1 \leq i \leq 2028$.

If the answer can be expressed in the form $\binom{x}{y}$, determine the values of x and y.

Answer:

Q16. A mysterious virus has corrupted Ben's Omnitrix, creating a distorted Omnitrix that overlaps the original one. The original Omnitrix is a circle of radius **r** where **only two sectors,each with a 90° central angle, are colored green** (the rest of the circle is black). The distorted Omnitrix is exactly as Omnitrix shown in the figure (same angle and radius as the original omnitrix). The power of the corrupted Omnitrix is defined as :

Power = | (Green-only Area + Orange-only Area) − Overlapping Red-only Area | (absolute value)

Where, Green-only Area is covered only by the original Omnitrix. Orange-only Area is covered only by the distorted Omnitrix. Overlapping Red-only Area is covered by both. Ben can defeat an alien if and only if Power > Alien Power

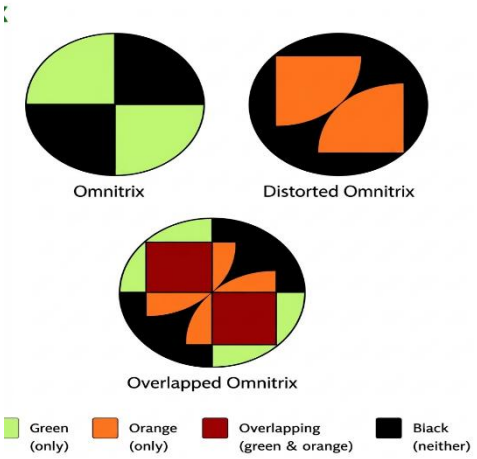


Figure 1. Original Omnitrix, Distorted Omnitrix, and their overlap.

If, the radius of the Omnitrix is 25, power of aliens 10 ,15 ,25 ,30 ,35, 50 ,60 , 80 ,100 . How many aliens Ben can defeat?

Answer:

MCQ Section (10 × 2)

Instructions: ***Each question carries 2 marks****. Choose the correct option by marking ✓the corresponding letter. There is only one correct answer per question****. 0.5 marks will be deducted for each wrong answer.***

Q1. $(((2^0)^1)^9)^{2020}$ =?

(a) 1 (b) 0 (c) 2030 (d) 2

Q2. How many complete floors (including basement) are there in 12 tola building? [hard]

(a) 8 (b) 9 (c) 10 (d) 12

Q3. 2024 × (-1+2024) × (-2+2024) × (-3+2024) × ..... (-4048+2024) =?

(a) 0 (b) 13524324167342 (c) 1524260298324 (d) 2224323452266

Q4. which one is a prime number?

(a)129 (b)13 (c)25 (d)25

Q5. The numbers from 1 to 20 are written on a board. One number is erased. The sum of the remaining numbers is 203. Which number was erased?

(a)12 (b)5 (c)7 (d)3

Q6. Given the integer 23423426, find the remainder when it is divided by 67

(a) 23 (b)24 (c)26 (d)25

Q7. If $-9 \leq x \leq 3$and $-4 \leq y \leq 10$, find the absolute difference between the maximum and minimum values of xy.

(a) 102 (b) 114 (c) 144 (d) 126

Q8. Shafik encrypted the following message using a Caesar cipher.

AIPGSQI XS TVSKVEQQMRK GSRXIWX

The shift value $K$is one of 2, 4, or 5. Exactly one value decrypts the message into a meaningful English phrase.

(a) Welcome to Programming Contest (b) Welcome to Programming Challenge (c) Welcome to Competitive Programming (d) Welcome to MathForces Contest

Q9. Determine the sum of all even numbers among the first 100 palindromic binary numbers

(a)0 (b)42 (c)42 (d)126

Q10. What is the Sum of first 15 prime numbers?

(a)328 (b)312 (c)428 (d)228